

BANDIRMA, Turkey

WMO: 171150

Lat: 40.3162N

Lon: 27.9723E

Elev: 42

StdP: 100.82

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.4	-2.0	-6.9	2.1	-1.3	-5.4	2.4	-0.2	18.2	1.3	15.2	4.5	2.9	0	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	32.2	20.6	30.8	20.6	29.5	20.8	23.7	28.0	23.0	27.3	22.4	26.9	5.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.0	26.3	21.7	16.4	26.0	20.9	15.6	25.7	71.3	28.0	68.5	27.4	66.3	27.2	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	11.5	10.3	-6.6	36.9	2.5	2.8	-8.4	38.9	-9.9	40.5	-11.3	42.1	-13.1	44.1
			-7.0	24.7	2.5	0.9	-8.8	25.4	-10.3	26.0	-11.7	26.5	-13.5	27.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	5.5	6.4	8.6	12.2	17.0	21.8	24.6	25.0	20.7	16.0	11.2	7.2
	DBStd	7.58	3.83	3.94	3.68	3.38	2.87	2.52	1.87	1.98	2.83	3.31	3.78	3.86
	HDD10.0	483	148	112	73	14	0	0	0	0	0	2	30	105
	HDD18.3	1941	399	333	302	185	60	3	0	0	11	86	217	345
	CDD10.0	2213	7	12	29	82	217	354	453	465	322	188	65	18
	CDD18.3	628	0	0	1	2	18	107	195	207	83	14	2	0
	CDH23.3	4992	0	0	3	32	198	853	1655	1673	499	71	7	0
	CDH26.7	1369	0	0	0	5	41	246	500	449	114	14	0	0
Wind	WSAvg	4.6	4.5	4.9	4.5	3.9	3.8	4.3	5.3	5.9	4.8	4.7	3.9	4.6
Precipitation	PrecAvg	725	112	75	72	62	37	27	16	18	34	65	94	126
	PrecMax	1086	334	212	169	212	122	108	84	118	154	205	363	349
	PrecMin	482	12	6	8	8	4	0	0	0	0	0	24	11
	PrecStd	142	76	45	43	39	29	28	21	28	39	48	60	68
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	19.8	22.9	27.0	30.2	34.2	35.1	34.0	32.2	28.2	24.2	19.4
		MCWB	12.1	13.0	14.3	17.0	18.7	20.0	20.7	21.0	19.5	18.0	16.4	13.8
	2%	DB	15.2	17.0	19.9	23.5	27.4	31.1	32.1	31.7	29.8	25.1	21.1	17.1
		MCWB	11.1	11.8	12.9	15.4	18.2	19.6	20.5	21.7	19.3	17.9	15.2	12.5
	5%	DB	13.2	15.0	17.8	21.2	25.7	29.2	30.5	30.1	27.4	23.0	19.0	15.1
		MCWB	9.9	10.8	12.0	14.4	17.4	19.7	20.8	21.9	19.5	17.3	14.3	11.6
	10%	DB	11.4	12.9	15.2	19.0	23.5	27.8	29.2	29.0	26.0	21.1	17.0	13.0
		MCWB	8.8	9.6	10.8	13.5	16.7	19.6	20.8	21.8	19.5	16.8	13.6	10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	13.6	15.0	17.7	20.4	23.2	24.1	24.8	22.9	20.2	17.6	14.4
		MCDB	16.5	18.2	21.8	25.5	27.1	27.6	28.7	29.1	26.3	24.0	22.1	18.4
	2%	WB	11.4	12.2	13.3	16.0	19.1	22.0	23.3	24.1	21.7	18.9	15.8	12.9
		MCDB	14.7	16.3	18.6	22.2	25.5	27.0	28.0	28.3	25.7	23.1	19.7	16.2
	5%	WB	10.2	10.9	12.3	14.9	18.1	21.2	22.7	23.4	21.0	18.1	14.8	11.8
		MCDB	13.1	14.4	17.1	20.5	23.9	26.3	27.4	27.6	25.2	22.1	18.1	14.8
	10%	WB	9.1	9.8	11.1	13.7	17.3	20.5	22.1	22.8	20.3	17.2	13.7	10.7
		MCDB	11.3	12.7	15.0	18.5	22.3	25.9	27.0	27.0	24.6	20.6	16.9	13.0
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	9.1	10.1	10.5	10.3	9.0	8.0	9.3	8.3	8.7	7.1
		MCDBR	10.8	12.3	14.5	15.6	15.6	14.3	12.2	10.2	13.0	12.4	12.5	9.8
		MCWBR	7.6	8.0	8.7	9.0	8.0	6.3	5.3	4.3	6.3	6.7	7.9	6.6
	5% WB	MCDBR	9.9	11.1	13.1	13.9	13.0	11.0	8.6	7.8	9.6	9.7	10.6	8.9
		MCWBR	7.1	7.4	8.2	8.4	7.1	5.2	4.2	3.6	5.2	5.7	7.0	6.3
Clear-Sky Solar Irradiance	taub	0.369	0.400	0.437	0.479	0.445	0.429	0.432	0.440	0.429	0.415	0.383	0.372	
	taud	2.323	2.216	2.120	2.028	2.203	2.292	2.293	2.268	2.283	2.289	2.324	2.316	
	Ebn at Noon	770	797	807	799	838	851	844	826	809	772	747	731	
	Edh at Noon	91	116	142	166	143	131	129	129	120	107	90	86	
All-Sky Solar Radiation	RadAvg	1.60	2.30	3.61	5.01	6.37	7.21	7.52	6.51	4.86	3.17	2.06	1.47	
	RadStd	0.22	0.31	0.35	0.56	0.47	0.50	0.25	0.29	0.34	0.29	0.26	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.42	N/A	+1.02	N/A	+0.64	+0.81	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+0.56	N/A	N/A	+0.58	N/A	N/A	N/A	-126	+137	+68

Nomenclature: See separate page